🔑 **Answer Key — Teacher Copy · Lesson 7-9**

MSTAR PRACTICE WORKSHEET

Lesson 7-9: Reflect Points Across Axes

Grade 6 Mathematics · Standard 6.NOS.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

What is the reflection of (4, -3) over the x-axis?

A. (-4, -3)

B. (-4, 3)

C. (3, -4)

D. (4, 3)

Reflecting over the x-axis changes the sign of the y-coordinate. The x stays the same (4), and the y changes from -3 to 3. The answer is (4, 3).

Question 2 SELECTED RESPONSE — CORRECT: B

What is the reflection of (-2, 6) over the y-axis?

A. (-2, -6)

B. (2, 6)

C. (2, -6)

D. (6, -2)

Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2, and y stays 6. The answer is (2, 6).

Question 3 SELECTED RESPONSE — CORRECT: B

Which point is on the y-axis?

A (-2, 4)

B (0, 5)

C (3, 3)

D (5, 0)

Points on the y-axis have x-coordinate = 0. (0, 5) is on the y-axis.

Question 4 SELECTED RESPONSE — CORRECT: C

Point A is at (-3, 5). It is reflected over the x-axis to A', then A' is reflected over the y-axis to A''. What are the coordinates of A''?

A (-3, -5)

B (-3, 5)

C (3, -5)

D (3, 5)

Over x-axis: (-3, 5) → (-3, -5). Then over y-axis: (-3, -5) → (3, -5). Both coordinates changed sign.

Question 5 SELECTED RESPONSE — CORRECT: D

Captain Vega folds the map along the x-axis. A marker at (7, 2) lands on its mirror image. What are the new coordinates?

A (-7, 2)

B (-7, -2)

C (2, 7)

D (7, -2)

Folding along (reflecting across) the x-axis flips the sign of the y-coordinate only. (7, 2) becomes (7, -2).

Question 6 SELECTED RESPONSE — CORRECT: C

A buried chest is marked at (-5, 3). Vega folds the map along the y-axis to find its mirror twin. Where is the twin?

A (-5, -3)

B (3, -5)

C (5, 3)

D (5, -3)

Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Reflect the point (5, −2) across the x-axis. What are the coordinates of the image?

A (−2, 5)

B (5, 2)

C (−5, −2)

D (−5, 2)

Reflecting across the x-axis changes the sign of the y-coordinate only: (x, y) → (x, −y), so (5, −2) → (5, 2).

- **A:** The coordinates were swapped. Keep x at 5 and change the sign of y.
- **C:** That reflects across the y-axis, which changes x. Reflecting across the x-axis changes y.
- **D:** Both signs were changed. Reflecting across the x-axis changes only the y-coordinate.

Part B — correct answer: B

Why does reflecting across the x-axis change only the y-coordinate?

A The x-coordinate is always positive after a reflection.

B The x-axis is horizontal, so the point flips vertically while its horizontal position stays fixed.

C Only the second coordinate can be negative.

D Reflections always change exactly one coordinate to zero.

A reflection across a line flips the point to the other side of that line at the same distance. Across the horizontal x-axis, only the vertical position changes.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Reflecting Across the Wrong Axis

> Dante reflected (3, 7) across the y-axis. He wrote: 'Change the sign of y, so the image is (3, –7).'

Explain Dante's misconception and give the correct image point.

> **Model answer:** (–3, 7). Reflecting across the y-axis (a vertical line) flips the point horizontally: (x, y) → (–x, y). Dante used the x-axis rule, (x, y) → (x, –y), which reflects across the horizontal axis instead.

2 points	Student identifies that a reflection across the y-axis changes the sign of the x-coordinate, gives (–3, 7), and explains that Dante applied the x-axis rule instead.
1 point	Student gives (–3, 7) without explaining which rule Dante used.
0 points	Student agrees with Dante or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A triangle has vertices at A(1, 2), B(4, 2), and C(4, 6). If you reflect the entire triangle over the y-axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

> **Model answer:** Reflect each vertex over the y-axis by flipping only the x-coordinate: A(1, 2) → A'(-1, 2), B(4, 2) → B'(-4, 2), and C(4, 6) → C'(-4, 6). The reflected triangle is congruent to the original — same side lengths and same angles — because a reflection is a mirror flip that turns the figure around without stretching or shrinking it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.